# Device5 - SHAP Feature Importance Comparison (All Features)

**Models**

| | HH_L1_weight | HH_L3_magnitude | HH_L3_weight | HH_L1_magnitude | HH_L5_magnitude | HH_L1_mean | HH_L5_weight | HH_L3_mean | HH_L5_mean | BigFeat_Gen_1 | BigFeat_Gen_0 | BigFeat_Gen_5 | BigFeat_Gen_2 | BigFeat_Gen_3 | BigFeat_Gen_4 | HH_L1_radius | HH_L1_std | HH_L3_radius | HpHp_L0.01_pcc | HH_L5_std | HH_L3_std |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ExtraTrees Original | 0.0231 | 0.0190 | 0.0194 | 0.0170 | 0.0164 | 0.0164 | 0.0149 | 0.0137 | 0.0146 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0080 | 0.0081 | 0.0066 | 0.0074 | 0.0066 | 0.0065 |
| ExtraTrees BigFeat | 0.0169 | 0.0103 | 0.0120 | 0.0106 | 0.0082 | 0.0093 | 0.0122 | 0.0105 | 0.0081 | 0.0153 | 0.0134 | 0.0112 | 0.0103 | 0.0115 | 0.0087 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| TPOT Original | 0.0238 | 0.0202 | 0.0152 | 0.0186 | 0.0196 | 0.0185 | 0.0143 | 0.0191 | 0.0173 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0080 | 0.0071 | 0.0072 | 0.0058 | 0.0065 | 0.0054 |
| TPOT BigFeat | 0.0208 | 0.0151 | 0.0134 | 0.0107 | 0.0116 | 0.0115 | 0.0130 | 0.0093 | 0.0093 | 0.0132 | 0.0133 | 0.0126 | 0.0130 | 0.0076 | 0.0102 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |

**Features (Sorted by Average Importance)**

SHAP Value

0.020
0.015
0.010
0.005
0.000